1) Voting: A voting algorithm selects the trajectory to execute from the approximated Pareto set $\mathcal{P}$ according to externally estimated operational risk indicators (Fig. 1, block (4)). Let $\mathbf{r} = [\mathcal{WR}, \mathcal{CR}, \mathcal{LR}, \mathcal{BR}] \in [0,1]^4$ denote normalized estimates of wind, communication, localization, and battery operational risks, respectively. A value of 0 corresponds to nominal operational risk, while 1 denotes a critical operating condition. As mentioned in [1], their computation is application- and platform-dependent and is outside the scope of this work. ARENA therefore treats them as exogenous inputs, which, in the power-line inspection application considered here, are based on wind speed, motor saturation, data transmission speed, GPS satellite availability, dilution of precision, battery level, and return-to-home distance.

The costs $f_{i,j}$ of every solution $i \in \mathcal{P}$ and objective $j \in \{T, S, E\}$ are first normalized over the current set, so that no offline reference is needed:

$$\hat{f}_{i,j} = \frac{f_{i,j} - \min_{l\in\mathcal{P}} f_{l,j}}{\max_{l\in\mathcal{P}} f_{l,j} - \min_{l\in\mathcal{P}} f_{l,j}} \in [0,1]. \tag{1}$$

The influence of each risk on each objective is described by the matrix $\mathbf{A} = [a_{j,r}]$, with rows T, S, E and columns $\mathcal{WR}$, $\mathcal{CR}$, $\mathcal{LR}$, $\mathcal{BR}$:

$$\mathbf{A} = \begin{bmatrix} -1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix}. \tag{2}$$

Wind, communication, and localization risks favor safety over time, wind and battery risks also favor energy, and the battery risk favors time over safety. Each column is scaled to a largest entry of 1, so that every risk has the same authority. The coefficients of the objectives are

$$k_j = \frac{k_{j,0} \prod_r (1-r)^{-\beta a_{j,r}}}{\sum_{m\in\{T,S,E\}} k_{m,0} \prod_r (1-r)^{-\beta a_{m,r}}}, \tag{3}$$

where $k_{j,0}$ are the nominal coefficients and β is a gain. By construction, $k_T + k_S + k_E = 1$ for any risks and $k_j = k_{j,0}$ under nominal conditions. For low risks, Eq. 3 reduces to a linear adjustment of the nominal coefficients,

$$k_j \approx k_{j,0}\Big(1 + \beta\big(\textstyle\sum_r a_{j,r} r - \sum_m k_{m,0} \sum_r a_{m,r} r\big)\Big), \tag{4}$$

while, as a risk approaches 1, the coefficient of the objective it influences most tends to 1: a critical wind, communication, or localization risk selects the safest trajectory of $\mathcal{P}$, and a critical battery risk the fastest one.

The selected trajectory minimizes the augmented weighted Tchebycheff scalarization of the normalized costs:

$$i^* = \arg\min_{i\in\mathcal{P}} \Big(\max_j k_j \hat{f}_{i,j} + \epsilon \sum_j k_j \hat{f}_{i,j} \Big), \tag{5}$$

where $\epsilon \ll 1$ breaks ties toward non-dominated solutions. Unlike weighted sums of costs or of ranks, which can only select part of the Pareto set, every Pareto-optimal solution is the unique minimizer of Eq. 5 for some coefficients [3], [4], so any trajectory of $\mathcal{P}$ can be reached by the risks. We use $k_{j,0} = 1/3$, $\beta = 2$, and $\epsilon = 10^{-3}$. Adjusting the objectives' influence from the risks in real time addresses a key challenge in multi-objective optimization [2]. The entries of $\mathbf{A}$ are specific to the power-line inspection problem considered in this paper and can be adapted to other UAV platforms and mission types.

REFERENCES

[1] [Reference `petit2024` of the paper's bibliography.]

[2] [Reference `hayes2022` of the paper's bibliography.]

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[4] K. Miettinen, *Nonlinear Multiobjective Optimization*. Boston, MA, USA: Kluwer, 1999.